

[Read about risk-groups here.](#)

Risk Group 1 (RG1; low individual and community risk)

A microorganism, nucleic acid, or protein that is either not capable of causing disease; or unlikely to do so. RG1 organisms pose a low risk to the health of individuals, and a low risk to public health. They maybe opportunistic and may pose a threat to immunocompromised individuals.

e.g.- *Escherichia coli*, *Saccharomyces cerevisiae*, *Recombinant protein*

Risk Group 2 (RG2; moderate individual risk, low community risk)

A pathogen or toxin that poses a moderate risk to the health of individuals or animals, and a low risk to public health. They are able to cause serious disease but are unlikely to do so. Effective treatment and preventive measures are available and the risk of spread of diseases caused by these pathogens is low.

e.g.- Human blood, most mammalian cell lines, *Streptococcus pyogenes*, Adenovirus, Lentivirus, *Staphylococcus aureus*.

Risk Group 3 (RG3; high individual risk, low community risk)

A pathogen that poses a high risk to the health of individuals but a low risk to public health. They are likely to cause serious disease. Effective treatment and preventive measures are usually available and the risk of spread of disease is low for the public. The risk of spread to the animal population, however, can range from low to high depending on the pathogen.

e.g.- *Mycobacterium tuberculosis*, Human Immunodeficiency virus, Creutzfeldt-Jakob disease (prion)

Risk Group 4 (RG4; high individual risk, high community risk)

A pathogen that poses a high risk to the health of individuals and a high risk to public health. They are likely to cause serious disease which can often lead to death. Effective treatment and preventive measures are not usually available and the risk of spread of disease caused by these pathogens is high for the public. The risk of spread of disease to the animal population, however, ranges from low to high depending on the pathogen.

e.g.- Ebola Virus, Marburg virus, Lassa virus

ePATHOgen database <https://health.canada.ca/en/epathogen>